

COURSE DESCRIPTION			
Program	Architecture, Interior Design		
Course Title	Architectural Graphics		
Course Code	1181	Semester	I
Type of Course	Drawing	Contact Hours	45
Teaching Scheme (L: T:P:R)	0:1:2:0	Credits	2
CIE Marks	40	ESE Marks	60

Introduction

The course equips students with fundamental architectural graphics skills to visually communicate design ideas through standard drawing conventions, projections, and basic building drawings.

Course Objectives

1	To familiarize the language and principles of architectural graphics used to communicate ideas, design intent, and construction information effectively.
2	To develop proficiency in the use of drawing instruments, graphical conventions, and dimensioning practices in accordance with relevant BIS standards through guided practice and self-learning exercises.
3	To develop the ability to interpret and generate orthographic views, sectional views, and developments of simple geometric solids and architectural objects using standard projection methods.
4	To enhance spatial visualization skills by converting between pictorial, orthographic, and isometric representations through structured drawing tasks and independent practice.
5	To apply graphical skills in producing single-line drawings of simple buildings and residential units, including plans, elevations, and sections drawn to scale.
6	To cultivate drawing discipline and presentation skills such as line quality, lettering, proportion, and layout for professional architectural documentation.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic Geometry of Secondary school level			Secondary Education

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Apply fundamental principles of architectural graphics and basic projection techniques to prepare elementary architectural drawings.	Apply
CO2	Apply the principles of orthographic projection to prepare accurate first-angle projection drawings of simple objects.	Apply
CO3	Apply isometric and perspective projection techniques to prepare two-dimensional drawings in accordance with standard drafting practices.	Apply
CO4	Apply standard architectural drawing conventions to prepare scaled single-line plans, elevations, and sections of simple buildings.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	1	-	-	1	-	-	-	2	-	2
CO2	3	2	1	1	-	-	-	-	2	-	1
CO3	3	2	2	1	1	-	-	-	2	-	1
CO4	3	2	3	1	1	1	1	-	3	-	2
Total	12	7	6	3	3	1	1	0	9	-	6
Correlation Level	3	1.75	2	1	1	1	1	0	2.25	-	1.5

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
0	1	2	0	2	45	2	40	60	100	0	0	0	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	Fundamentals of Architectural Graphics	Drawing Instruments & Line Conventions - Dimensioning methods - Geometrical constructions and Curves - Scales & Introduction to Projections	3	6
Module II	Orthographic projection & Sectional views	Orthographic views of pointed solids and prisms - Conversion of pictorial views into orthographic views - Sectional view of objects	4	8
Module III	Isometric and Perspective projections	Introduction to isometric projections - Perspective projections- One point perspective projection - Perspective projections- Two point perspective projection	4	8
Module IV	Single line drawings of buildings	Single line plan of a single room - Elevation & Section - Single room - Single line plan of a single-storey residence - Residential Elevation, Section & Presentation	4	8

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours	Action
Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours	
Module I						
Drawing Instruments & Line Conventions	Lecture: Identify drawing instruments used in architectural graphics and the importance of standard line conventions in drawings.	T	CO1	Remember	0.5	
Drawing Instruments & Line Conventions	Experiment: Identification and correct use of drawing instruments. Practice of standard line types as per BIS on drawing sheets.	P	CO1	Apply	1	
Dimensioning methods	Lecture: Identify principles of dimensioning, elements of dimensioning, and BIS recommendations.	T	CO1	Remember	0.5	
Dimensioning methods	Experiment: Apply aligned and unidirectional dimensioning systems on simple geometric	P	CO1	Apply	1	
Geometrical Constructions and Curves	Lecture: Identify basic geometrical constructions and commonly used curves in architectural drawings.	T	CO1	Remember	1	
Geometrical Constructions and Curves	Experiment: Construct perpendicular bisector, division of line, regular polygons; draw ellipse (concentric circle method) and parabola (focus-directrix method).	P	CO1	Apply	2	
Scales & Introduction to Projections	Lecture: Explain types of scales, RF, architectural scales, and introduce the concept of projection and projection planes.	T	CO1	Understand	1	
Scales & Introduction to Projections	Experiment: Construct plain and diagonal scales and practice basic projections of points and straight lines- parallel, and perpendicular to HP & VP, parallel to both planes (use first angle projection only)	P	CO1	Apply	2	
Module II						

Orthographic views of pointed solids and prisms	Lecture: Explain principles of orthographic projection and first-angle projection method.	T	CO2	Understand	1
Orthographic views of pointed solids and prisms	Experiment: Draw orthographic views (plan, elevation, side view) of prisms and pointed solids using first-angle projection.	P	CO2	Apply	2
Conversion of pictorial views into orthographic views	Lecture: Explain the method of visualizing and converting pictorial views into orthographic views.	T	CO2	Understand	2
Conversion of pictorial views into orthographic views	Experiment: Develop orthographic views of simple objects containing plain surfaces, slanting surfaces, slots, cylindrical surfaces. (use First Angle Projection method only)	P	CO2	Apply	4
Sectional view of objects	Lecture: Explain sectional planes, full section and half section concepts.	T	CO2	Understand	1
Sectional view of objects	Experiment: Draw full and half sectional views of objects by selecting appropriate cutting planes.	P	CO2	Apply	2
Module III					
Introduction to isometric projections	Lecture: Explain isometric scale, isometric view, and isometric projection with examples.	T	CO3	Understand	1
Introduction to isometric projections	Experiment: Construct isometric scale and draw isometric view of rectangular block.	P	CO3	Apply	2
Perspective projections- One point perspective projection	Lecture: Explain terminology and principles of perspective projection with emphasis on one-point perspective.	T	CO3	Understand	1
Perspective projections- One point perspective projection	Experiment: Draw one-point perspective of an interior room	P	CO3	Apply	2
Perspective projections- Two point perspective projection	Lecture: Explain application of perspective projection in architectural drawings with focus on two-point perspective.	T	CO3	Understand	2
Perspective projections- Two point perspective projection	Experiment: Draw two-point perspective views of simple solids.	P	CO3	Apply	4
Module IV					
Single-Line plan of a Single Room	Lecture: Explain principles of single-line drawings, symbols, and scaling.	T	CO4	Understand	1
Single-Line plan of a Single Room	Experiment: Draw a single-line plan of a single-room building to scale.	P	CO4	Apply	2
Elevation & Section - Single Room	Lecture: Explain coordination between plan, elevation, and section.	T	CO4	Understand	1
Elevation & Section - Single Room	Experiment: Draw elevation and section for the single-room building.	P	CO4	Apply	2
Single line plan of a Single-storey residence	Lecture: Explain planning principles of a simple residence.	T	CO4	Understand	1
Single line plan of a Single-storey residence	Experiment: Draw a single-line plan of a single-storey residential building to scale.	P	CO4	Apply	2
Residential Elevation, Section & Presentation	Lecture: Explain presentation techniques, line quality, lettering, and sheet organisation.	T	CO4	Understand	1
Residential Elevation, Section & Presentation	Experiment: Draw elevation and section of the residence and organize drawings into a neat presentation sheet.	P	CO4	Apply	2

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Prepare a line convention chart with examples. Online quiz for familiarisation of drawing instruments	2
2	Practice conventional lines with names and applications Apply aligned and unidirectional dimensioning methods on simple figures.	2
3	Construct regular polygons such as heptagon or octagon	2

4	Draw two straight lines perpendicular to each other and construct an arc of radius 50 mm tangential to both the lines. Take photos of real buildings showing curves (arch, parabolic roof) and annotate digitally.	2
Module II		
5	Identify plan, elevation and side view of plane figures.	1
6	Draw orthographic views of the pyramid and cylinder.	1
7	Make a paper model of a simple solid and develop its orthographic views (plan, elevation, side view)	2
8	Revise Orthographic Projections	2
Module III		
9	Construct an isometric scale. Draw isometric view of any architectural feature	3
10	Convert orthographic views into isometric views of simple objects. Exercise to improve line quality and dimensioning	4
11	Identify perspective terminology and elements through labelled sketch. Draw one point perspective of a bedroom	3
12	Draw birds eye view of a city by watching tutorials	1
Module IV		
13	Identify various symbols used in architectural drawings. Draw single line plan of a bedroom to scale	3
14	Draw a single line plan of a residence	1
15	Arrange drawings into a neat portfolio for submission.	1
Total Self-Learning hours		30

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Architectural Graphics	Francis D. K. Ching	John Wiley & Sons
2	Engineering Graphics	P I Varghese	VIP Publishers

REFERENCE			
Sl. No.	Title	Author	Publication
1	Engineering Graphics and Design	A.P. Gautam, Pradeep Jain, Ankita Maheshwari	Khanna Publishing House
2	Engineering Drawing	N D Bhatt	Charotar Publishing House
3	Engineering Drawing	R K Dhawan	S. Chand & Company
4	Engineering Drawing	Dhananjay Jolhe	Tata McGraw Hill
5	Engineering Drawing	Shah P. J	
6	Fundamentals of Drawing	K.R.Gopalakrishna	Subhas Publications
7	Geometrical Art and Drawing	I. H.Morris	

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://www.slideshare.net/slideshow/architectural-drawings-47711760/47711760#2	I
2	https://www.slideshare.net/slideshow/lesson-13perspectiveprojection/27520803	III

Assessment Methodology - Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)

Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	40	40	60
Converted to		15	15	15	15	20	20	
Marks	5	15				20		60
Total	40							60
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

First Internal Examination Pattern of Question Paper - Drawing

Duration	Modules	TYPE OF QUESTIONS				
		PART A		PART B		Total Mark
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (7 marks each)	
2 hrs	Module 1	5	5	2	14	19
	Module 2	0	0	3	21	21
	Total		5		35	40

Second Internal Examination Pattern of Question Paper - Drawing

Duration	Modules	TYPE OF QUESTIONS				
		PART A		PART B		Total Mark
		No. of Questions	Marks (7 marks each)	No. of Questions	Marks (13 marks each)	
2 hrs	Module 3	2	14	0	0	14
	Module 4	0	0	2	26	26
	Total		14		26	40

End Semester Examination Pattern of Question Paper for Drawing

Duration	Module	TYPE OF QUESTIONS						
		PART A (1 out of 2 questions from each Module)		PART B		PART C		Total Marks
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (8 marks each)	No. of Questions	Marks (15 marks each)	
2.5 hrs	1	5	5	1/2*	8	0	0	13
	2	0	0	2/3*	16	0	0	16
	3	0	0	2/3*	16	0	0	16
	4	0	0	0	0	1/2*	15	15
	Total	5	5	5	40	1	15	60
x/y* Represents x number of questions to be answered from the given y number of questions.								

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